CS2 Role Classification Using Unsupervised Learning

Background

This project analyzes professional Counter-Strike 2 demo files to discover player roles using unsupervised machine learning. Instead of training a supervised model on predefined labels, the system attempts to determine whether recognizable player archetypes emerge naturally from gameplay behavior alone.

The pipeline processes raw .dem replay files(demos) from top-tier professional tournaments, engineers behavioral and positional features, applies multiple clustering algorithms, and evaluates the resulting role structures using both traditional clustering metrics and a custom bootstrap stability framework.

Professional Counter-Strike provides an interesting environment for this type of analysis because player roles are partially structured but rarely rigidly defined. While some roles, such as dedicated AWPers, are relatively easy to recognize statistically, others depend heavily on positioning, utility usage, timing, or team coordination. This makes role discovery a challenging machine learning problem with substantial ambiguity and noise.

The project currently uses professional match demos from two S-tier events: IEM Atlanta 2026, PGL Astana 2026. I plan to add games from CS Asia Championship 2026, and IEM Cologne Major. It is unlikely that new data outside of these four tournaments would be added, as the roster locks for the IEM Cologne Major create a unique opportunity where team rosters do not change. The dataset consists of 176 demos covering 320 professional players.

The project was implemented entirely in Python using awpy for demo parsing, Polars and Pandas for data processing, scikit-learn for clustering and evaluation, HDBSCAN for density-based clustering, XGBoost and RandomForest for interpretability analysis, and Matplotlib for visualization generation.

Problem Statement

Professional Counter-Strike teams are often described in terms of player roles. Analysts, commentators, and fans frequently refer to players as entry fraggers, lurkers, anchors, support players, AWPers, and in-game leaders (IGLs). However, these roles are rarely rigidly defined. Many players shift responsibilities depending on team structure, economy state, or map strategy, making manual classification subjective and inconsistent.

This project investigates whether meaningful player archetypes can emerge directly from gameplay data using unsupervised learning rather than predefined labels. The core question is:

> Can professional CS2 player roles be discovered from behavioral and positional statistics?

Pipeline

The project uses an automated data pipeline that transforms raw professional Counter-Strike 2 match demos into clustered player role profiles.

The pipeline begins with automated extraction and deduplication of professional match demos downloaded manually from HLTV.org. A custom extraction script scans recent downloads, extracts valid .dem replay files, removes duplicates, and moves new demos into the project dataset directory.

Once extracted, demos are processed through a multiprocessing-based parser built on top of the awpy library. Awpy allows the parser to extract a wide range of gameplay information including combat events, utility usage, opening duels, trade interactions, movement behavior, and positional relationships between players.

A large feature engineering stage transforms this raw data into behavioral statistics describing how players fight, move, coordinate with teammates, and use utility. Features are normalized by rounds played, and both averages and standard deviations are recorded to capture variability. The final dataset stores one row per player per side (CT and T), allowing side-specific behaviors to be analyzed independently.

The pipeline then evaluates multiple unsupervised clustering algorithms including KMeans, Gaussian Mixture Models (GMM), and HDBSCAN. Each clustering configuration is evaluated using silhouette score, Davies-Bouldin index, and a custom bootstrap stability analysis based on Adjusted Rand Index (ARI). The stability analysis repeatedly resamples the dataset and recomputes clusters to measure how consistently discovered structures reappear. A ground truth label comparison is also used using liquipedia.net player role labels.

PCA scatter plots, radar charts, stability bar charts, silhouette-versus-k plots, and z-score heatmaps are created to assist with analysis. These visualizations make it possible to identify separable role structures, detect outlier players, and understand which gameplay characteristics define each cluster.

Feature Engineering

Feature engineering was a major part of the project. Since player roles in professional Counter-Strike are not directly labeled, clustering quality depended heavily on building features that captured meaningful differences in playstyle rather than just raw performance. The parser extracts information from combat events, utility usage, movement, and player positioning. Instead of relying only on standard statistics such as ADR or kills per round, the dataset includes metrics intended to describe how players take space, interact with teammates, use utility, and engage opponents. All features are normalized by rounds played, and both averages and standard deviations are recorded to capture behavioral consistency as well as central tendency.

Combat features describe damage output and weapon specialization, including ADR, kills per round, survival rate, damage differential, multi-kill rate, and kill share broken down by rifle and AWP. These metrics help separate players with different combat styles and levels of weapon use, as an AWPer and an entry fragger may have similar ADR but may look completely different in kill share and damage differential.

Opening duel metrics measure early-round aggression and entry tendencies through opening kill rate, opening death rate, opening duel success rate, and first contact rate. Players who consistently make first contact with the enemy show a distinct behavioral signature from players who may play more conservatively before engaging.

Trading metrics capture how players operate within team structure. A trade kill is defined as a kill that occurs within five seconds of a teammate's death against the player who made that kill. Trade kill rate, death traded rate, and trade participation together help distinguish coordinated support players from lurkers and independent fraggers who operate in isolation.

Utility metrics record grenade usage rates, flash assists, utility damage, smoke usage, and incendiary usage per round. These features are particularly important for identifying players whose primary contribution comes from enabling teammates rather than direct combat output, which proved critical for recovering the IGL cluster on T side.

Finally, positional metrics are computed directly from replay tick data and describe where players spend their time relative to teammates and opponents. These include average distance to enemies, distance to the team centroid, nearest teammate distance, movement distance per round, time spent near enemies, and stationary behavior rate. Positional features were the dominant drivers of cluster separation on the CT side, where the distinction between players is expressed more through spacing than through combat statistics.

Clustering Methodology

KMeans, Gaussian Mixture Models (GMM), and HDBSCAN were each included because they capture different types of structure within the data. KMeans partitions players into compact clusters based on centroid distance, while GMM models clusters probabilistically, allowing softer boundaries between overlapping playstyles. HDBSCAN was included because it can identify outliers naturally without forcing every player into a cluster, making it useful for detecting elite specialists or highly flexible players whose behavior differed substantially from the rest of the dataset.

Before clustering, all numeric features were standardized using StandardScaler so that features with larger numeric ranges would not dominate the clustering process. A Principal Component Analysis (PCA) was also computed for visualization purposes, allowing high-dimensional player profiles to be projected into two dimensions for interpretation.

Models were ranked by a composite score combining silhouette score (40%), an inverted Davies-Bouldin index (20%), and bootstrap stability (40%), where the stability term was penalized by its own standard deviation to discount models whose cluster assignments varied substantially across resamples. Silhouette score measures how well-separated clusters are, while Davies-Bouldin index measures cluster compactness and overlap. However, these metrics alone can produce misleading results in high-dimensional spaces, especially in noisy behavioral data.

To address this, the project implements a bootstrap stability analysis based off of Adjusted Rand Index (ARI). For each clustering configuration, the dataset is repeatedly resampled and reclustered. The resulting cluster assignments are then compared against the original clustering solution using ARI in order to measure how consistently the discovered structures reappear across perturbations of the data.

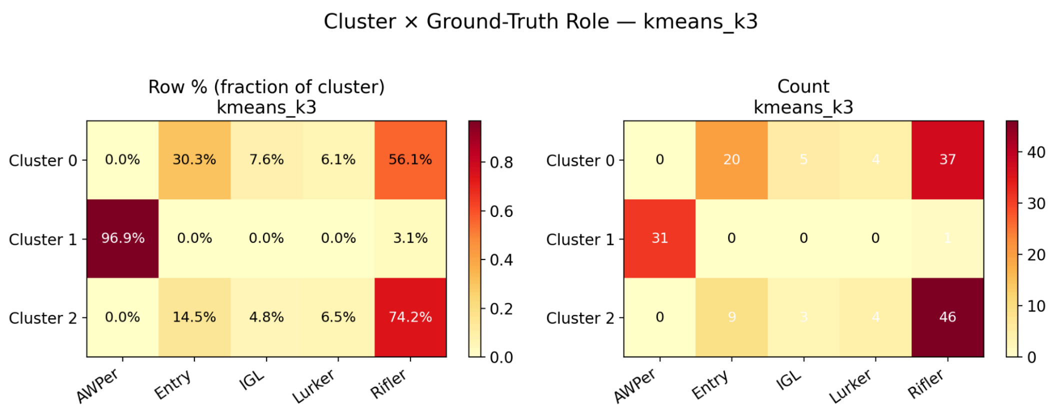
Results and Analysis

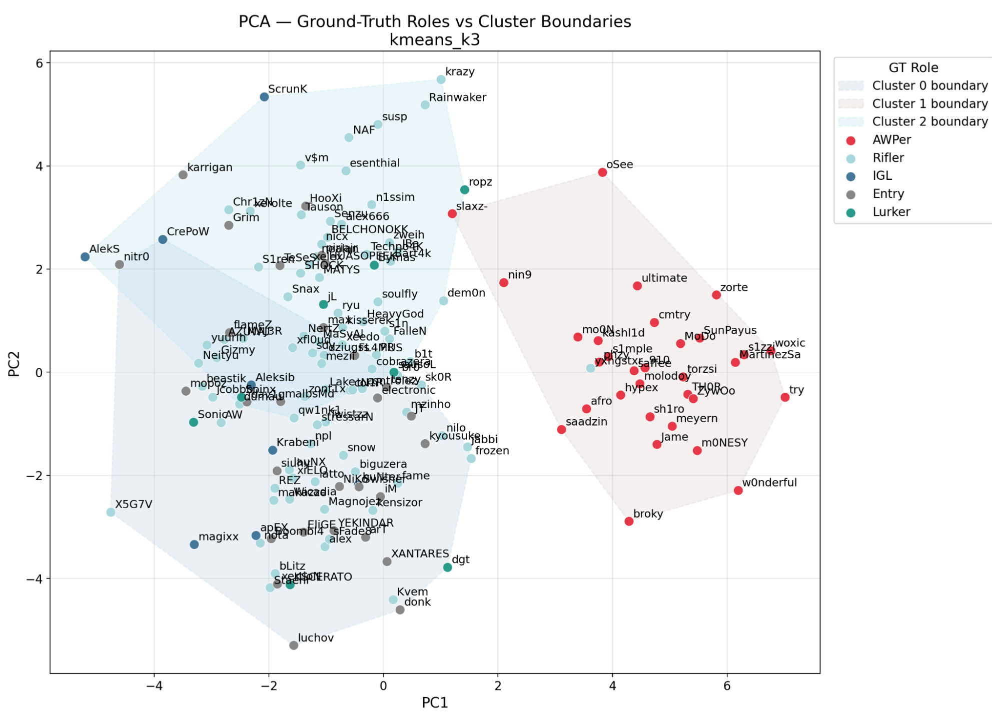
CT Side

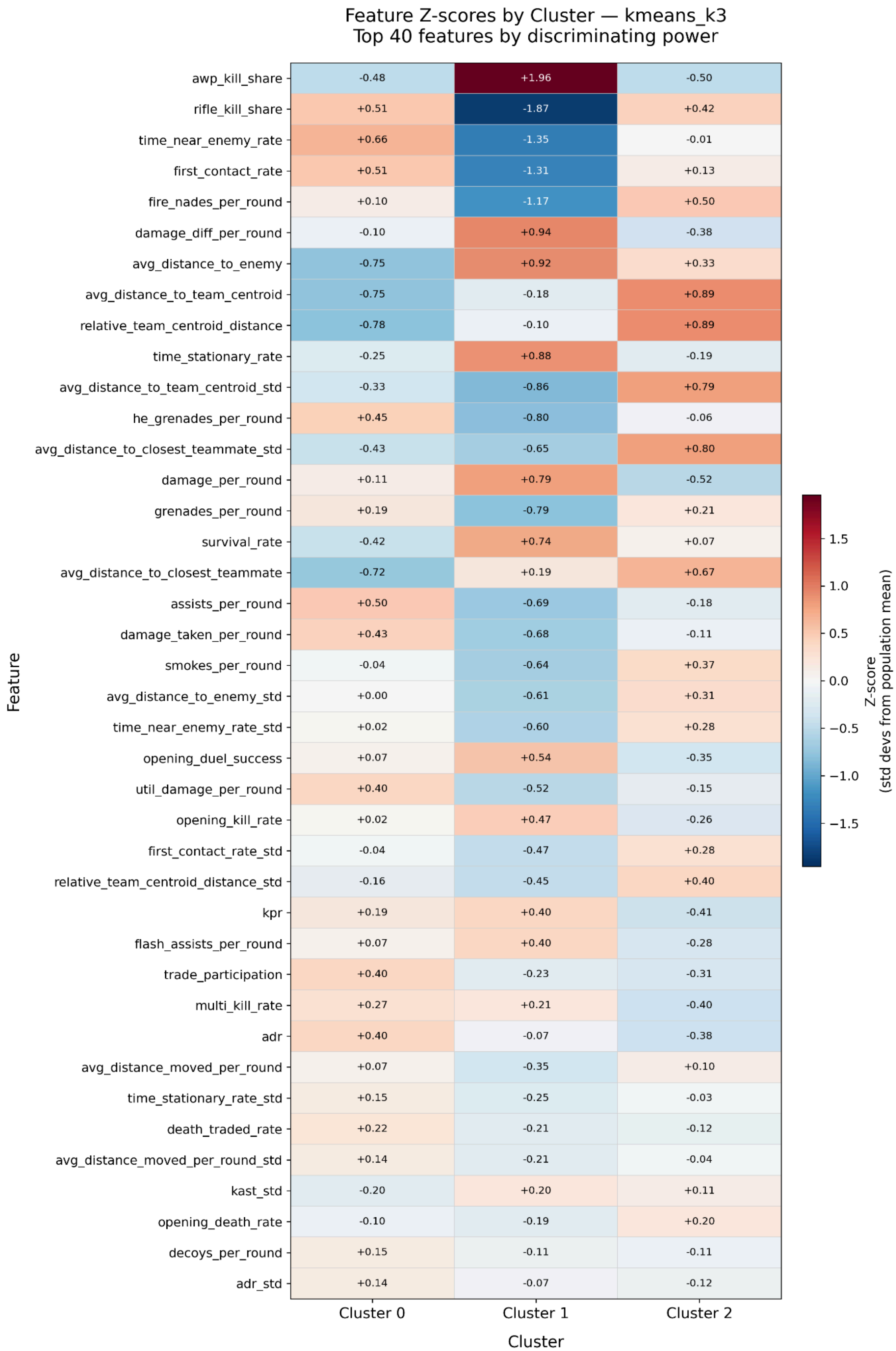
The CT side is best described by three clusters.

Cluster 1 - AWPers (96.9% purity): Near-perfect recovery, consistent with the T side result. The cluster shows an extreme positive z-score on AWP kill share (+1.96) with a strongly negative rifle kill share (-1.87). Notably, AWPers on CT do not play significantly further from their team than other players, suggesting that on CT the AWP is used more to hold close angles reactively than to take isolated positions.

Clusters 0 and 2 - Riflers (56.1% and 74.2% purity): Two rifler clusters emerge with a clear split between them. Cluster 2 sits substantially further from the team centroid and nearest teammate, while Cluster 0 plays closer to teammates but shows higher aggression metrics including first contact rate and time near enemy rate. The model is separating passive anchor-type riflers from more aggressive contact-seeking ones, though neither maps cleanly to a named role since Entry, Lurker, IGL, and Support players are all absorbed into these two groups.

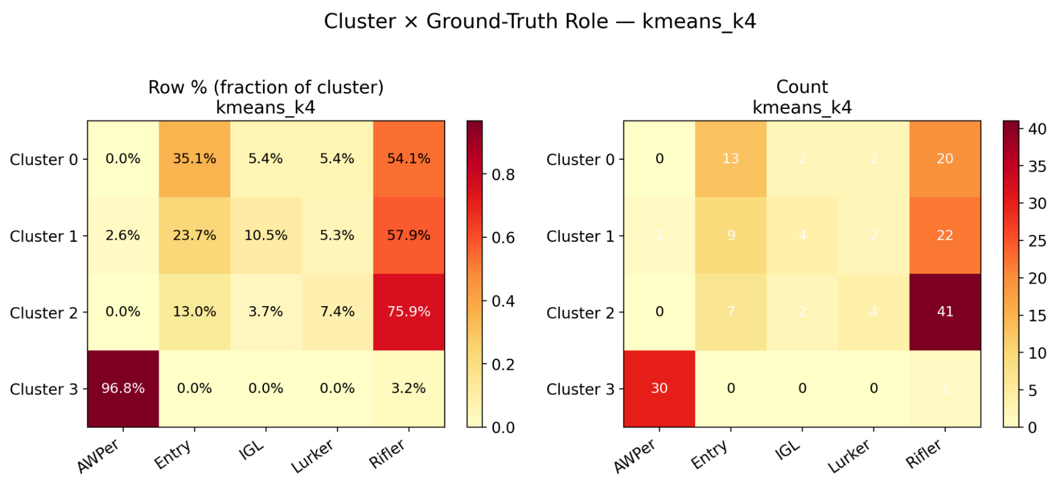






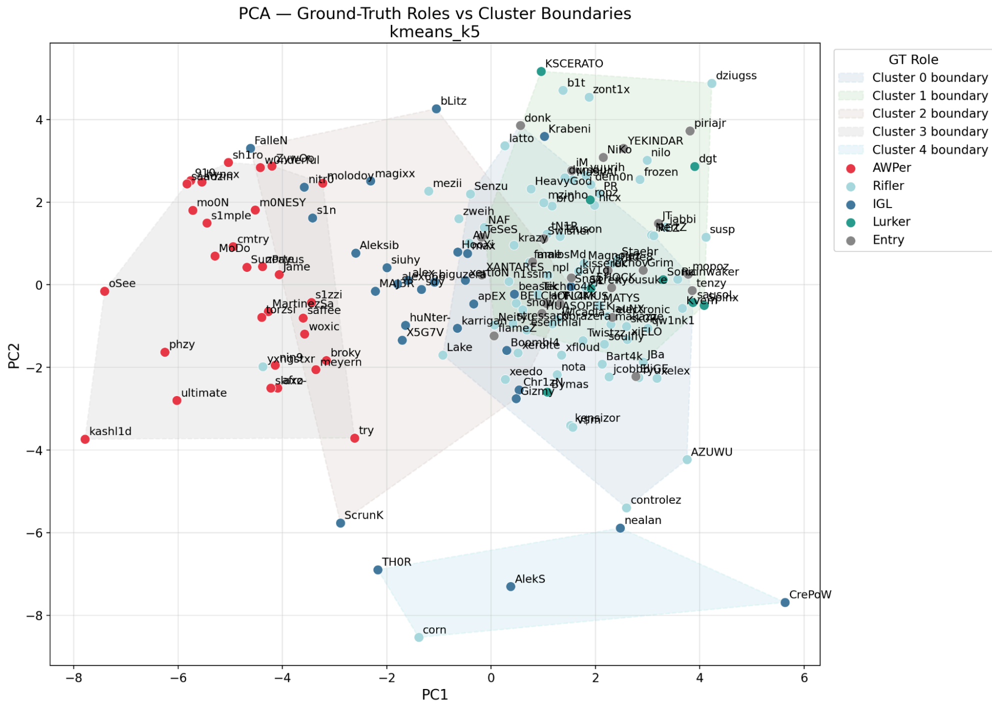
Feature Z-scores by Cluster — kmeans_k3
Top 40 features by discriminating power
awp_kill_share -0.48 +1.96 -0.50
rifle_kill_share +0.51 -1.87 +0.42
time_near_enemy_rate +0.66 -1.35 -0.01
first_contact_rate +0.51 -1.31 +0.13
fire_nades_per_round +0.10 -1.17 +0.50
damage_diff_per_round -0.10 +0.94 -0.38
avg_distance_to_enemy -0.75 +0.92 +0.33
avg_distance_to_team_centroid -0.75 -0.18 +0.89
relative_team_centroid_distance -0.78 -0.10 +0.89
time_stationary_rate -0.25 +0.88 -0.19
avg_distance_to_team_centroid_std -0.33 -0.86 +0.79
he_grenades_per_round +0.45 -0.80 -0.06
avg_distance_to_closest_teammate_std -0.43 -0.65 +0.80
damage_per_round +0.11 +0.79 -0.52
grenades_per_round +0.19 -0.79 +0.21
survival_rate -0.42 +0.74 +0.07
avg_distance_to_closest_teammate -0.72 +0.19 +0.67
assists_per_round +0.50 -0.69 -0.18
damage_taken_per_round +0.43 -0.68 -0.11
smokes_per_round -0.04 -0.64 +0.37
avg_distance_to_enemy_std +0.00 -0.61 +0.31
time_near_enemy_rate_std +0.02 -0.60 +0.28
opening_duel_success +0.07 +0.54 -0.35
util_damage_per_round +0.40 -0.52 -0.15
opening_kill_rate +0.02 +0.47 -0.26
first_contact_rate_std -0.04 -0.47 +0.28
relative_team_centroid_distance_std -0.16 -0.45 +0.40
kpr +0.19 +0.40 -0.41
flash_assists_per_round +0.07 +0.40 -0.28
trade_participation +0.40 -0.23 -0.31
multi_kill_rate +0.27 +0.21 -0.40
adr +0.40 -0.07 -0.38
avg_distance_moved_per_round +0.07 -0.35 +0.10
time_stationary_rate_std +0.15 -0.25 -0.03
death_traded_rate +0.22 -0.21 -0.12
avg_distance_moved_per_round_std +0.14 -0.21 -0.04
kast_std -0.20 +0.20 +0.11
opening_death_rate -0.10 -0.19 +0.20
decoys_per_round +0.15 -0.11 -0.11
adr_std +0.14 -0.07 -0.12
Feature
Cluster 0
Cluster 1
Cluster 2
Cluster
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Z-score
(std devs from population mean)

The key limitation on the CT side is that the Liquipedia role taxonomy is primarily T-side oriented. Entry and Lurker in particular describe T-side responsibilities that have no direct CT equivalent, making ground-truth comparison inherently noisy on this side. Moving to kmeans_k4 adds a fourth cluster but does not recover a new role archetype, merely subdividing the existing rifler mass with similar purity (54%, 57.8%, 75.9%).

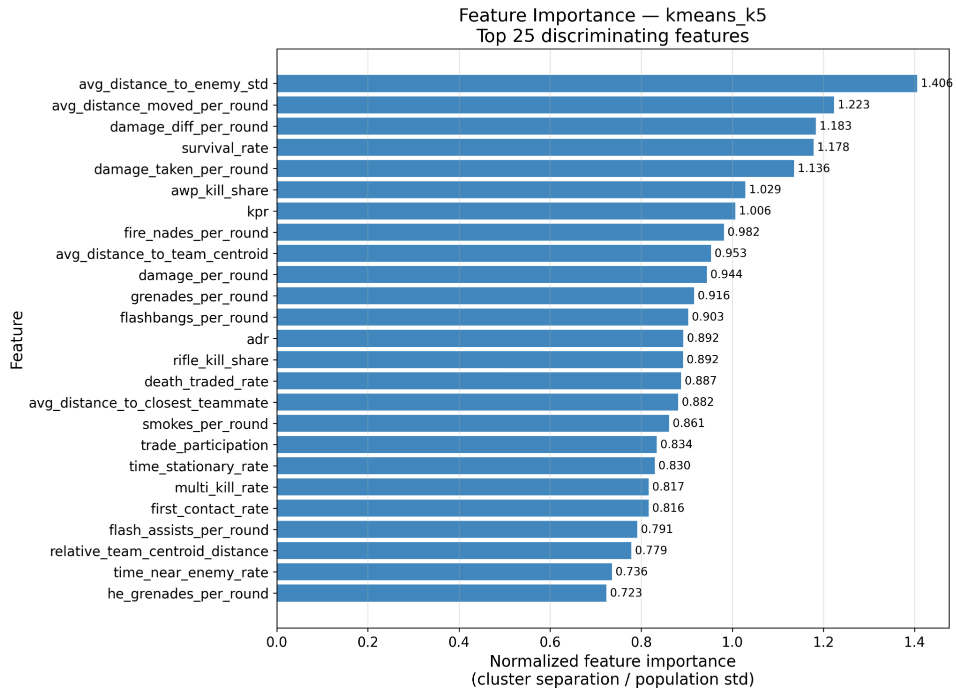


T Side

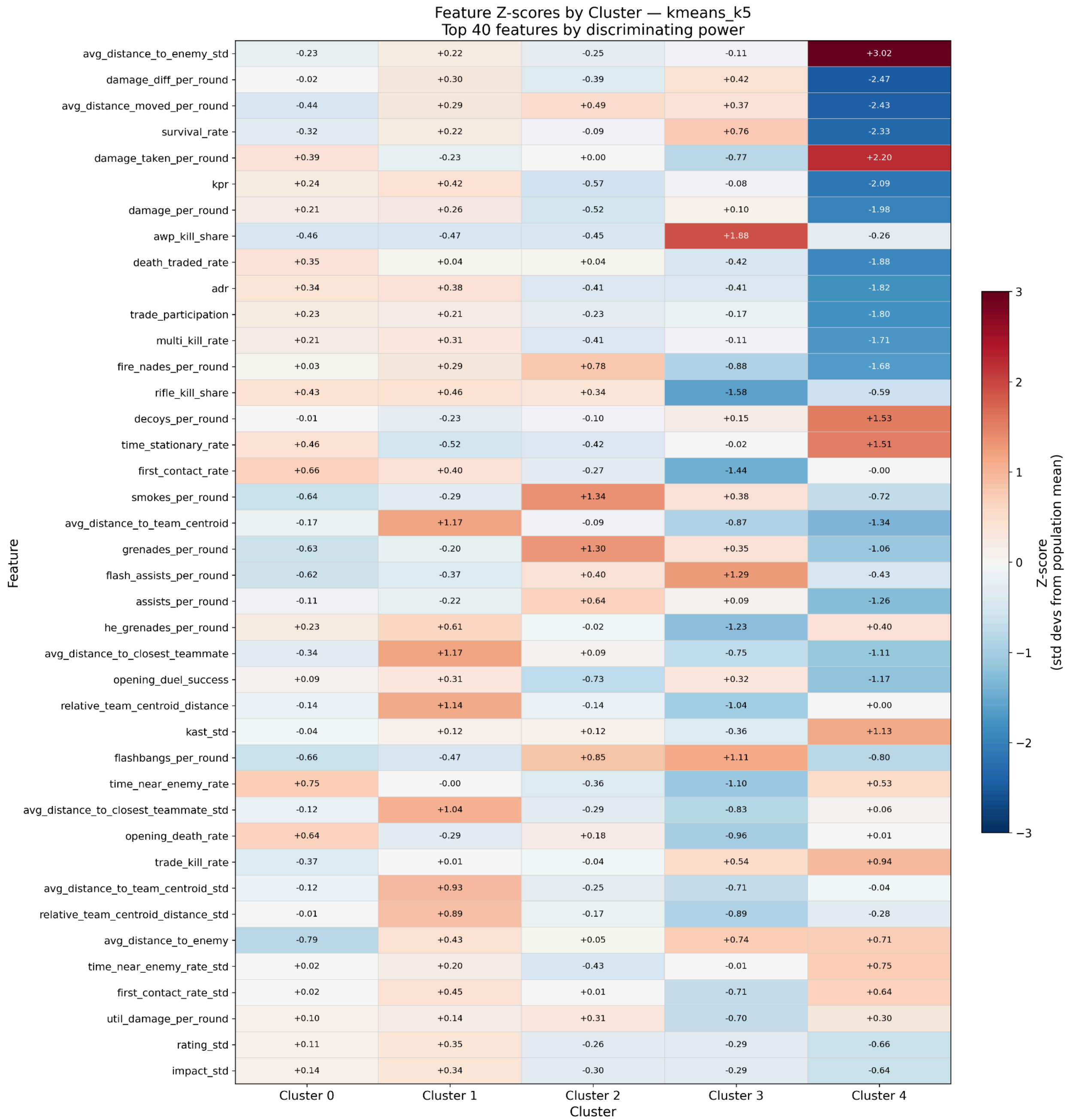
The T side is substantially more structured than CT. kmeans_k5 achieves 77.5% overall purity and an ARI of 0.421, significantly higher than the CT side's ARI of 0.291 which suggests that T-side roles are more behaviorally distinct than CT-side roles. This also makes sense intuitively, as on the T side, players are more flexible and proactive, while CTs tend to have to react to the moves of the T sided players. Five clusters emerge with interpretable identities:



Clusters 0 and 1 - Riflers (67.9% and 69.2% purity): Two rifler clusters emerge, both with similar purity. As on the CT side, the split is also along an aggression axis, where one cluster plays more aggressively (higher first contact rate, higher average distance to enemy) and one plays more supportively (higher trade participation, higher survival rate). The 67–69% purity on these clusters means roughly a third of their members have non-Rifler ground-truth labels, primarily Entry and Lurker players whose statistics overlap with these riflers.

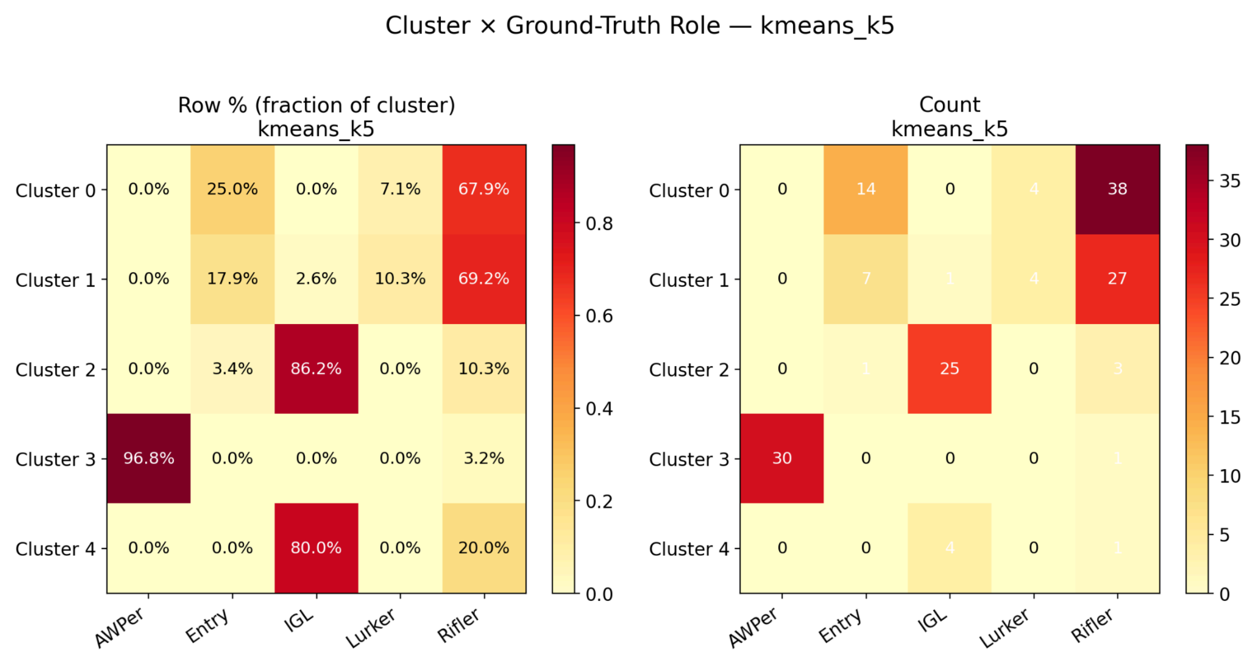


Cluster 2 - IGLs (86.2% purity): This is the most surprising result in the entire project. IGLs were expected to be the hardest role to detect because their defining contribution, calling and strategy decisions, generates no directly observable signal. Yet the model recovers them at 86.2% purity. The feature importance table explains why: on the T side, utility metrics, such as smokes per round, flashbangs per round, grenades per round, and flash assists per round all rank highly. IGLs execute more utility per round, flash more often for teammates, and use more smokes. Average distance to closest teammate is also a top feature, suggesting IGLs tend to play closer to their team rather than take isolated positions.



Cluster 3 - AWPers (96.8% purity): Near-perfect recovery, consistent with all other models across both sides. Similarly to the CT side clusters, AWP kill share dominates as the top feature. This cluster is essentially noise-free.

A fifth cluster of only 5 players also returned an IGL-dominant label at 80% purity. These players belong to Fisher College, a collegiate team whose Liquipedia role labels are inaccurate. Their substantial skill gap relative to the rest of the dataset caused the model to isolate them into a separate cluster rather than absorb them into the broader rifler groups.

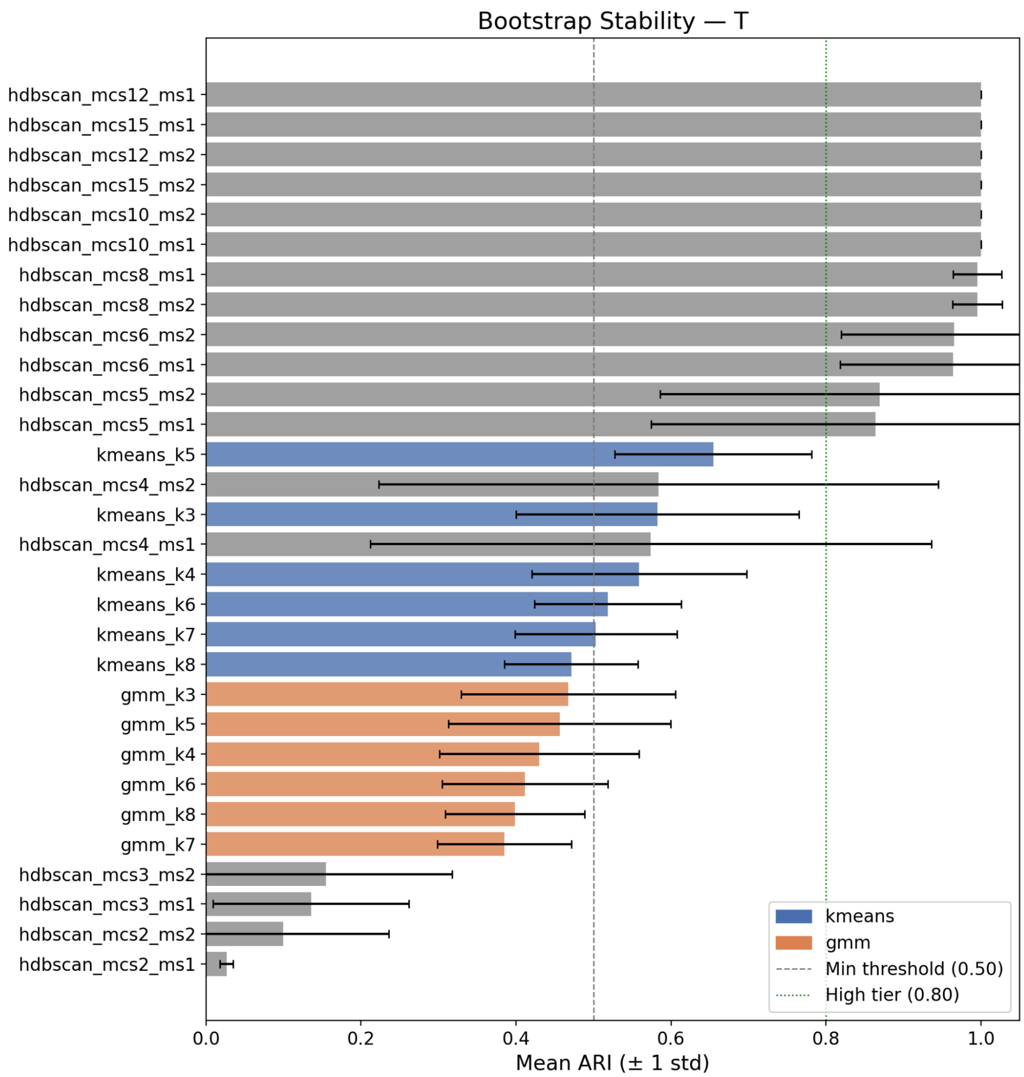


The higher ARI on the T side (0.421 vs 0.291) and the IGL recovery are the two most significant findings in the project. The IGL result in particular suggests that in-game leadership on T side has a measurable behavioral signature in utility usage and teammate proximity, even without any direct signal from voice communication or strategy calls.

Cross Model Observations

Across every algorithm, every k value, and on both sides, AWPers form a clean, stable cluster. This is not a surprising finding on its own, but it has an important implication: AWP kill share is so dominant that it crowds out other structures at low k values. For example, at k=3, the model essentially learns AWPer and everyone else rather than anything else.

CT side has less recoverable structure than T side. The ARI gap (0.291 CT vs 0.421 T) and the failure to recover any non-AWPer role with high purity on CT suggest that CT roles are either more fluid, more context-dependent, or insufficiently captured by the current feature set.



Surprisingly, GMM underperformed KMeans on stability across both sides. Every GMM configuration fell below the 0.50 stability threshold. This is likely because GMM's soft probabilistic boundaries are less consistent under resampling when the underlying clusters are not Gaussian, which professional player distributions, which are relatively consistent to each other, may not be.

HDBSCAN is not suitable for this dataset. Despite perfect bootstrap stability (ARI=1.0) on both the T and CT side, HDBSCAN's stability was artificial. It produced the same degenerate two-cluster solution across the entire hyperparameter grid, while labeling a large proportion of players as outliers. Professional players operate in a uniformly high-skill regime without the sharp density discontinuities, which HDBSCAN requires to form meaningful clusters beyond the AWPer group.

XGBoost Analysis

To validate that the discovered clusters represented stable and learnable behavioral structures rather than arbitrary partitions, XGBoost classifiers were trained using the cluster assignments as labels and evaluated on a held-out test set of 25% of the players. Feature importance was additionally computed using Random Forest classifiers to identify which gameplay features most strongly defined each player archetype.

XGBoost Classification Report — T kmeans_k5

Cluster	Precision	Recall	F1-Score	Support
Cluster 0	0.92	0.86	0.89	14
Cluster 1	0.89	0.80	0.84	10
Cluster 2	0.62	0.71	0.67	7
Cluster 3	1.00	1.00	1.00	8
Cluster 4	0.50	1.00	0.67	1
Accuracy			0.85	40
Macro Avg	0.79	0.87	0.81	40
Weighted Avg	0.87	0.85	0.85	40

On the T side, kmeans_k5 achieved 85% overall accuracy and a weighted F1 of 0.85, confirming that the five clusters captured actual behavioral structure. The AWPer cluster was classified perfectly, with a precision, recall, and F1 of 1.00, consistent with its near-perfect purity against Liquipedia labels. The two rifler clusters achieved F1 scores of 0.89 and 0.84, with occasional misclassifications between them. The IGL cluster achieved an F1 of 0.67, the weakest among the interpretable clusters, which is expected, given that IGL behavior overlaps with support-oriented riflers in feature space. Cluster 4, the Fisher College outlier cluster, had only one test player and its score is not meaningful.

XGBoost Classification Report — CT kmeans_k3

Cluster	Precision	Recall	F1-Score	Support
Cluster 0	0.89	1.00	0.94	17
Cluster 1	1.00	1.00	1.00	8
Cluster 2	1.00	0.87	0.93	15
Accuracy			0.95	40
Macro Avg	0.96	0.96	0.96	40
Weighted Avg	0.96	0.95	0.95	40

On the CT side, kmeans_k3 achieved 95% overall accuracy and a weighted F1 of 0.95, higher than the T side despite CT having less recoverable role structure. AWPers are perfectly separated at F1 1.00, and the two rifler clusters score 0.94 and 0.93. The spatial split between aggressive players and more passive anchor-style players is clear enough that the classifier reproduces it reliably, even though both clusters share the same ground-truth label. The high CT accuracy reflects cluster learnability rather than role richness, since both non-AWPer clusters map to the same ground-truth role.

Across every model, AWP kill share was consistently the most important feature, with rifle kill share also ranking highly, confirming that weapon specialization is the dominant source of role separation in the current feature space. On CT side models, positional metrics such as distance to team centroid, nearest teammate distance, and average distance to enemies were among the strongest predictors, suggesting CT-side differences are driven more by spacing than combat output. On T side models, utility features became substantially more important, with smokes per round, flash assists, and grenade usage ranking highly in the k=5 model, directly supporting the IGL cluster recovery. The HDBSCAN models showed far less variation in feature importance, with AWP kill share dominating nearly every configuration, further confirming that HDBSCAN was recovering only the most statistically extreme players rather than meaningful role structure.

Limitations

The most significant limitation is the sample size. 160 players is small for unsupervised learning, and the dataset is further constrained to a narrow performance tier, all players are top-level professionals. This compresses the feature distributions and makes it harder for any clustering method to find structure beyond AWPers.

The ground-truth labels from Liquipedia are also imperfect. Role assignments reflect community consensus and primary role designation, but many players legitimately fill multiple roles depending on map, economy, and team structure. A player labeled "Rifler" who also calls strategies half the time will appear in both the rifler and IGL clusters without either assignment being wrong. The CT side compounds this further, as the Liquipedia terminology is primarily T-side oriented, as roles like Entry and Lurker describe T-side responsibilities with no direct CT equivalent, meaning CT clusters are being evaluated against labels that were never designed to describe CT behavior. Players may also change their roles throughout their career, and as a community-edited wiki, Liquipedia role labels may not always be updated for less popular players.

Finally, the current feature set is side-averaged across all maps. Map-specific behavior, such as which bombsite a player anchors, which T-side role they take on particular maps is averaged away. Adding map-stratified features or per-map clustering would likely recover additional structure, particularly on the CT side where positional play is more map-dependent. However, this would require a much larger sample size, as a team may only play some maps several times in an entire tournament

Conclusion

This project demonstrated that meaningful player role archetypes can emerge from professional CS2 gameplay data using unsupervised learning, without any predefined labels. While AWPers were recovered reliably across every configuration, the more significant finding was that in-game leaders formed a statistically distinct cluster on the T side at 86.2% purity, suggesting that IGL behavior has a measurable signature in utility usage and teammate proximity that the model can detect without any direct signal from strategy or communication. The most natural extension of this work would be to incorporate map-stratified features, which would allow the model to separate positional behavior that is currently averaged across maps and potentially recover additional structure on the CT side.